

NONPARAMETRIC SEQUENTIAL PROCEDURES FOR SELECTING THE BEST OF k POPULATIONS

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1. Introduction

Let $\pi_1, \pi_2, \dots, \pi_k$ denote k populations ($k \geq 2$) with distribution functions $F(x - \Delta_1), F(x - \Delta_2), \dots, F(x - \Delta_k)$ where F is symmetric about 0, so that $\Delta_1, \dots, \Delta_k$ are the centres of symmetry of the respective distributions. The function F and the parameters Δ_i ($i = 1, \dots, k$) are considered to be unknown and we assume without loss of generality that $\Delta_1 \leq \Delta_2 \leq \dots \leq \Delta_k$. If $\Delta_k > \Delta_{k-1}$, π_k is referred to as the best population.

The problem considered in this paper is that of selecting the best population with probability at least P^* whenever $\Delta_k - \Delta_{k-1} \geq \Delta^*$, where P^* and Δ^* are preassigned constants with $1/k < P^* < 1$ and $\Delta^* > 0$. In other words, if we denote by CS a correct selection, by Ω the set of all vectors $\omega = (\Delta_1, \dots, \Delta_k)$ with $\Delta_k - \Delta_{k-1} \geq \Delta^*$ and by $H_0(J)$ some subclass of the symmetric distributions (so that the members of $H_0(J)$ satisfy certain regularity conditions needed), we wish to obtain a procedure for which $P(\text{CS}) \geq P^*$ for all $\omega \in \Omega$ and all $F \in H_0(J)$.

Robbins, Sobel and Starr (1968) and Srivastava (1966) considered sequential procedures for this problem. The procedure of Robbins, Sobel and Starr takes at each stage of the sequential sampling one observation from each population. Denote the observations taken up to the n -th stage by $X_{ij}, i = 1, \dots, k; j = 1, \dots, n$ and define

$$\bar{X}_i(n) = \sum_{j=1}^n X_{ij}/n, i = 1, \dots, k \text{ and}$$

$$u_n = (k(n-1))^{-1} \sum_{i=1}^k \sum_{j=1}^n (X_{ij} - \bar{X}_i(n))^2.$$

the procedure can then be described as follows: Terminate sampling at the N -th stage, where N is the least integer $n \geq 5$ such that $n \geq u_n h^2 / (\Delta^*)^2$, where h is the constant defined in (5.1). If β is the smallest j such that $\bar{X}_j(N) \geq \bar{X}_1(N)$ for $i = 1, \dots, k$, select π_β . It is proved that as $\Delta^* \rightarrow 0, N \rightarrow \infty$ a.s., and that

$$(1.1) \quad \lim (EN) (\Delta^*)^2 = \sigma^2 h^2$$

$$(1.2) \quad \liminf P(CS) \geq P^* \text{ for } \omega \in \Omega$$

provided $0 < \sigma^2 < \infty$. Certain small sample properties are considered also.

Sometimes it might be desirable to base the selection of the best population on statistics other than the sample means. Recently Geertsema (1972) considered two sequential procedures where, instead of the sample mean, the usual Hodges-Lehman estimator and the sample median, respectively, were used. In this paper we propose a sequential procedure which is based on the general Hodges-Lehmann estimator (to be defined below) and consider asymptotic properties of the procedure as $\Delta^* \rightarrow 0$, obtaining results analogous to (1.1) and (1.2).

Sen and Ghosh (1971) studied a procedure based on a class of one-sample rank order statistics for obtaining a fixed-width confidence interval for the median of a symmetric distribution. In sections 2–4 we adopt their notation and make use of results analogous to theirs; this in preparation for studying the properties of the procedure we introduce in section 5 for selecting from k distributions the one with the largest center of symmetry.

2. Class of one-sample rank order statistics

Our procedure for determining the stopping variable N is based on the following class of one-sample rank order statistics:

Let X_{ij} denote the j -th observation taken from population π_i for $i=1, \dots, k$ and $j=1, 2, \dots$. Let $c(u) = 0$ or 1 according as $u < 0$ or not. Define

$$R_{n\alpha}(i) = \sum_{\beta=1}^n c(|X_{i\alpha}| - |X_{i\beta}|);$$

$$\alpha = 1, \dots, n; \quad \underline{X}_{in} = (X_{i1}, X_{i2}, \dots, X_{in})$$

$$(2.1) \quad T_n(i) = T(\underline{X}_{in}) = n^{-1} \sum_{\alpha=1}^n c(X_{i\alpha}) J_n(R_{n\alpha}(i)/(n+1)) \quad i=1, \dots, k$$

where

$$(2.2) \quad J_n(u) = EJ(U_{ni}) \text{ for } (i-1)/n < u \leq i/n, i=1, \dots, n$$

and $U_{n1} < U_{n2} < \dots < U_{nn}$ are the n ordered random variables from the rectangular $(0,1)$ df. Also, we assume that $J(u) = \psi^{-1}((1+u)/2)$, $0 \leq u < 1$, where $\psi(x)$ is an absolutely continuous cdf defined on $(-\infty, \infty)$ satisfying the conditions

$$(2.3) \quad \psi(-x) + \psi(x) = 1, \text{ i.e., } \psi'(x) = r(x) = r(-x) \text{ for all real } x,$$

$$(2.4) \quad r(x)/[1-\psi(x)] \text{ is non-decreasing for all } x \geq x_0$$

where $x_0 (\geq 0)$ is some real number.

We denote by H_0 the class of all absolutely continuous df's G symmetric about 0 for which both the density function g and its first derivative g' exist and are bounded for almost all x . Also, let

$$(2.5) \quad H_0(J) = \{G : G \in H_0 \text{ and } \lim_{x \rightarrow \infty} g(x)J'(G(x) - G(-x)) \text{ is finite} \}.$$

Throughout this paper it will be assumed that $F \in H_0(J)$. We now introduce the following notation:

$$(2.6) \quad \bar{E}_n = n^{-1} \sum_{i=1}^n J_n(i/(n+1)); A_n^2 = n^{-1} \sum_{i=1}^n J_n^2(i/(n+1))$$

$$(2.7) \quad \mu = \int_0^1 J(u) du \quad \text{and} \quad A^2 = \int_0^1 J^2(u) du.$$

From the definition of the function $J_n(u)$ and (2.6) it now follows that $\bar{E}_n = \mu$ for all n . Define for $i=1, \dots, k$ and $-\infty < a < \infty$

$$(2.8) \quad W_n^{(i)}(a) = n^{1/2} [T(\underline{X}_{in}) - T(\underline{X}_{in} - a \underline{1}_n) - aB(F)]$$

where $\underline{1}_n = (1, \dots, 1)$ and

$$(2.9) \quad B(F) = \int_0^\infty (d/dx)J[2F(x) - 1]dF(x).$$

Also, let

$$(2.10) \quad g_r(n) = n^{-1/2}(\log n)^r, r \geq 1.$$

Theorem 2.1

If $\Delta_i = 0$ it follows from (2.3), (2.4) and (2.5) that

$$|a| \leq g_r(n) |W_n^{(i)}(a)| = O(n^{-1/2}(\log n)^{2r}) \quad a.s.$$

Thus,

$$|a| \leq g_r(n) |W_n^{(i)}(a)| \rightarrow 0 \quad a.s. \text{ as } n \rightarrow \infty$$

Proof:

See Sen and Ghosh (1971) theorem 4.3 page 195.

Under slightly less restrictive conditions as in (2.3), (2.4) and (2.5) Puri and Sen (1969) proved the following theorem:

Theorem 2.2

If $\Delta_i = 0$ then

$$\lim_{n \rightarrow \infty} P\{2n^{1/2}[T_{n^*}(i) - \mu/2]/A \leq x\} = \Phi(x) \text{ for all } i=1, \dots, k$$

where Φ is the c.d.f. of the standard normal distribution.

3. General Hodges-Lehmann estimator

Define for $i=1, \dots, k$

$$\begin{aligned} \Delta_i^* &= \Delta_i^*(X_{i1}, \dots, X_{in}) = \sup \{a : T(X_{in} - a \mathbf{1}_n) > \mu/2\} \\ (3.1) \quad \Delta_i^{**} &= \Delta_i^{**}(X_{i1}, \dots, X_{in}) = \inf \{a : T(X_{in} - a \mathbf{1}_n) < \mu/2\}. \end{aligned}$$

Now, let

$$(3.2) \quad \hat{\Delta}_i(n) = (\Delta_i^* + \Delta_i^{**})/2$$

then $\hat{\Delta}_i(n)$ is referred to as the general Hodges-Lehmann estimator of the parameter Δ_i . (See Hodges and Lehmann (1963) page 600).

Remarks:

- (i) $\hat{\Delta}_i(n)$ is translation invariant.
- (ii) The distribution function of $\hat{\Delta}_i(n)$ is absolutely continuous (see theorem 1 of Hodges and Lehmann (1963)).
- (iii) If ψ is uniform over $(-1, 1)$ then $\hat{\Delta}_i(n)$ is the median of the $n(n+1)/2$ averages $(X_{ij} + X_{is})/2$ for $j \leq s$; $j, s=1, \dots, n$ (see Hodges and Lehmann (1963) page 603). The procedure of Geertsema (1972) is based on this estimator.

Lemma 3.1

If $\Delta_i = 0$

$$\Delta_i^* = O(n^{-1/2}(\log n)^2) \quad \text{a.s.}$$

and $\Delta_i^{**} = O(n^{-1/2}(\log n)^2) \quad \text{a.s.}$

Proof:

It follows immediately from lemma 5.1 of Sen and Ghosh (1971) and the

Borel-Cantelli lemma.

Theorem 3.1

$$\hat{\Delta}_i(n) = \Delta_i + [T(\underline{X}_{in}) - \mu/2]/B(F) + O(n^{-3/4}(\log n)^6) \quad a.s.$$

Proof:

In order to prove the theorem, it is sufficient to prove

$$(i) \quad \Delta_i^* = \Delta_i + [T(\underline{X}_{in}) - \mu/2]/B(F) + O(n^{-3/4}(\log n)^6) \quad a.s.$$

and

$$(ii) \quad \Delta_i^{**} = \Delta_i + [T(\underline{X}_{in}) - \mu/2]/B(F) + O(n^{-3/4}(\log n)^6) \quad a.s.$$

We prove (i) only, since (ii) follows in the same way. Without loss of generality set $\Delta_i = 0$. From the definition of Δ_i^* (see 3.1) it follows that

$$(3.3) \quad T(\underline{X}_{in} - (\Delta_i^* + 1/n)\underline{1}_n) \leq \mu/2 \quad \text{and}$$

$$(3.4) \quad T(\underline{X}_{in} - (\Delta_i^* - 1/n)\underline{1}_n) > \mu/2.$$

From lemma 3.1 and theorem 2.1 we get

$$(3.5) \quad W_n^{(i)}(\Delta_i^* + 1/n) = O(n^{-3/4}(\log n)^6) \quad a.s.$$

i.e.

$$(3.6) \quad [T(\underline{X}_{in}) - T(\underline{X}_{in} - (\Delta_i^* + 1/n)\underline{1}_n) - \Delta_i^*B(F)] \\ = O(n^{-3/4}(\log n)^6) \quad a.s. \quad (\text{from (2.8)}).$$

Using (3.3) and (3.6) we obtain

$$(3.7) \quad \Delta_i^* \geq [T(\underline{X}_{in}) - \mu/2]/B(F) + O(n^{-3/4}(\log n)^6) \quad a.s.$$

Using (3.4) we get in the same way as above

$$(3.8) \quad \Delta_i^* \leq [T(\underline{X}_{in}) - \mu/2]/B(F) + O(n^{-3/4}(\log n)^6) \quad a.s.$$

Remarks:

The result of theorem 3.1 is a useful result which is analogous to Bahadur's

representation for sample quantiles in terms of the empirical distribution function (see Behadur (1966)). For instance, it follows immediately from theorem 3.1 together with theorem 2.2 that

$$(3.9) \quad \lim_{n \rightarrow \infty} P\{2n^{1/2}(\hat{\Delta}_i(n) - \Delta_i)B(F)/A \leq x\} = \Phi(x)$$

for $i=1, \dots, k$. We also note (although this will not be used in the present paper) that from Sen (1970) and theorem 3.1 we may conclude that

$$\hat{\Delta}_i(n) \rightarrow \Delta_i \quad \text{a.s.} \quad \text{as } n \rightarrow \infty.$$

Theorem 3.2

The sequence $\{\hat{\Delta}_i/n\}$ is uniformly continuous in probability with respect to $\{n^{-1/2}\}$ i.e., for every positive ϵ and $\eta > 0$, there exist a $\delta > 0$, such that as $n \rightarrow \infty$

$$P\{\sup_{|n'-n| < \delta n} |n^{1/2}(\hat{\Delta}_i(n') - \hat{\Delta}_i(n))| > \epsilon\} < \eta.$$

Proof:

It follows from theorem 3.1 that

$$n^{1/2}(\hat{\Delta}_i(n') - \hat{\Delta}_i(n)) = n^{1/2}[T(X_{in}') - T(X_{in})]/B(F) +$$

$$O(n^{-1/2}(\log n)^6) \quad \text{a.s.}$$

if $\Delta_i = 0$. The proof of the theorem now follows in exactly the same way as the proof of lemma 5.3 of Sen and Ghosh (1971).

4. A strongly consistent estimator of $[B^2(F)]^{-1}$

Let α ($0 < \alpha < 1$) be a constant. In a similar manner as (3.1) and (3.2) of Sen and Ghosh (1971) we define

$$(4.1) \quad \hat{\theta}_{1n}(i) = \sup\{a : T(X_{in} - a1_n) > T_{n\alpha}^{(1)}\}$$

$$(4.2) \quad \hat{\theta}_{2n}(i) = \inf\{a : T(X_{in} - a1_n) < T_{n\alpha}^{(2)}\} \quad (i=1, \dots, k)$$

where $T_{n\alpha}^{(1)}$ and $T_{n\alpha}^{(2)}$ are two known constants such that

$$P(T_{n\alpha}^{(2)} \leq T(X_{in}) \leq T_{n\alpha}^{(1)}) \rightarrow 1 - \alpha \quad \text{as } n \rightarrow \infty$$

if $\Delta_i = 0$. Let $\Phi(\tau_{\alpha/2}) = 1 - \alpha/2$ and define

$$(4.3) \quad S_n^2 = (n/(k\tau_{\alpha/2}^2 A^2)) \sum_{i=1}^k (\hat{\theta}_{2n}(i) - \hat{\theta}_{1n}(i))^2.$$

Using lemma 5.2 of Sen and Ghosh (1971) and the Borel-Cantelli lemma it follows that

$$(4.4) \quad S_n^2 \rightarrow [B^2(F)]^{-1} \text{ a.s. as } n \rightarrow \infty.$$

5. Definition of the Sequential Procedure

Let h denote the solution of

$$(5.1) \quad \int_{-\infty}^{\infty} \Phi^{k-1}(y+h) d\Phi(y) = P^*.$$

Tables of h for various values of P^* and k are given in Bechhofer (1954). The definition of the procedure (of which the motivation is the same as that given by Geertsema (1972)) is as follows:

At every stage of the sequential sampling take one independent observation from each population. Let the stopping variable N be the first $n \geq n_0$ such that

$$n \geq h^2 S_n^2 A^2 / 4(\Delta^*)^2$$

where n_0 is a suitable positive integer. Upon termination of sampling, select π_j if $\hat{\Delta}_j(N) \geq \hat{\Delta}_i(N)$ for all $i=1, \dots, k$.

Theorem 5.1

For all $\omega \in \Omega$ and $F \in H_0(J)$

- (a) $P(N < \infty) = 1$
- (b) $N \uparrow \infty$ a.s. as $\Delta^* \downarrow 0$, $E(N) \uparrow \infty$ as $\Delta^* \downarrow 0$
- (c) $N/\nu \rightarrow 1$ a.s. as $\Delta^* \rightarrow 0$
- (d) $E(N)/\nu \rightarrow 1$ as $\Delta^* \rightarrow 0$
- (e) $\liminf P(CS) \geq P^*$ as $\Delta^* \rightarrow 0$,

where $\nu = \nu(\Delta^*, P^*) = A^2 h^2 / 4(\Delta^*)^2 B^2(F)$.

Proof:

(a), (b) and (c) follow from the definition of N , (4.4), the monotone convergence theorem and Lemma 1 of Chow and Robbins (1965). To prove (d), it is sufficient to prove that the set $\{N/\nu\}$ for $\Delta^* > 0$ is uniformly integrable (see Loève (1963)). For this, it is sufficient to prove (see Bickel and Yahav (1968)) that

$$\sum_{n=1}^{\infty} 0 < \Delta^* < \sup_{\Delta^*} P(N/\nu > n) < \infty \text{ for some } \Delta^*_0 > 0.$$

Let $m = m_n(\nu)$ denote the largest integer less than or equal to $n\nu$. Thus,

$$\begin{aligned}
 & P(N > n\nu) \\
 & \leq P(N > m) \\
 & \leq P(S_m > 4(\Delta^*)^2 m/h^2 A^2) \\
 & = P\left(\sum_{i=1}^k (\theta_{2m(i)} - \hat{\theta}_{1m(i)})^2 > k \tau_{\alpha/2}^2 4(\Delta^*)^2/h^2\right) \\
 & \leq \sum_{i=1}^k P((\theta_{2m(i)} - \hat{\theta}_{1m(i)})^2 > 4(\Delta^*)^2 \tau_{\alpha/2}^2/h^2) \\
 & \leq \sum_{i=1}^k P((\theta_{2m(i)} - \hat{\theta}_{1m(i)}) > 2\Delta^* \tau_{\alpha/2}/h) \\
 & = \sum_{i=1}^k P(m^{1/2}(\theta_{2m(i)} - \hat{\theta}_{1m(i)}) > m^{1/2} 2\Delta^* \tau_{\alpha/2}/h)
 \end{aligned}$$

For any constant $\lambda > 1$ and for all $n \geq \lambda^2$ we have

$$\begin{aligned}
 & P(m^{1/2}(\theta_{2m(i)} - \hat{\theta}_{1m(i)}) > m^{1/2} 2\Delta^* \tau_{\alpha/2}/h) \\
 & \leq P(m^{1/2}(\theta_{2m(i)} - \hat{\theta}_{1m(i)}) > A \tau_{\alpha/2} \lambda/B(F)) \\
 & \leq c_s m^{-s} \text{ for all } m \geq m_s \quad (\text{independent of } \Delta^*)
 \end{aligned}$$

and $m \geq n\nu - 1 \geq \nu - 1 \geq m_s$ for $\Delta^* \leq \Delta^*_0$ and all n , where s is a constant greater than one. This follows from Lemma 5.2 of Sen and Ghosh (1971). Also, $m > n\nu(\Delta^*_0, P^*)$ for $\Delta^* < \Delta^*_0$; thus

$$\begin{aligned}
 & P(m^{1/2}(\theta_{2m(i)} - \hat{\theta}_{1m(i)}) > m^{1/2} 2\Delta^* \tau_{\alpha/2}/h) \\
 & \leq K n^{-s} \text{ where } K \text{ is a constant independent of } \Delta^*.
 \end{aligned}$$

This proves (d) of the theorem.

(e) Let $Y_i(n) = (2n)^{1/2}(\hat{\Delta}_k(n) - \hat{\Delta}_i(n) - (\Delta_k - \Delta_i))B(F)/A$, $i=1, \dots, k-1$.

The joint distribution of $Y_i(n)$, $i=1, \dots, k-1$ is asymptotically as $n \rightarrow \infty$ a multivariate normal distribution with mean vector $\bar{Q}$ and covariance matrix $\Sigma' = (\sigma'_{ij})$ with $\sigma'_{ii} = 1$, $\sigma'_{ij} = 1/2$, $1 \leq i, j \leq k-1$. This follows using (3.9) and the independence of $\hat{\Delta}_1(n), \dots, \hat{\Delta}_k(n)$. We now have

$$\begin{aligned}
P(CS) &= P(\hat{\Delta}_k(N) > \hat{\Delta}_i(N), i=1, \dots, k-1) \\
&= P(Y_i(N) > -(2N)^{1/2}(\Delta_k - \Delta_i)B(F)/A, i=1, \dots, k-1) \\
&\geq P(Y_i(N) > -(2N)^{1/2}\Delta^*B(F)/A, i=1, \dots, k-1)
\end{aligned}$$

for all $\omega \in \Omega$. By (c) of the present theorem, $N/\nu \rightarrow 1$ in probability and

$$(2N)^{1/2}\Delta^*B(F)/A \rightarrow h2^{-1/2} \text{ a.s. as } \Delta^* \rightarrow 0.$$

From theorem 3.2 it follows that the sequence $\{Y_i(n)/n^{1/2}\}$ is uniformly continuous in probability with respect to $\{n^{-1/2}\}$ for $i=1, \dots, k-1$. Thus, it follows from a result of Anscombe (1952) that

$$(5.2) \quad \liminf P(CS) \geq P(Z_i > -h2^{-1/2}, i=1, \dots, k-1)$$

as $\Delta^* \rightarrow 0$, where $Z_1, \dots, Z_{k-1}$ have a multivariate normal distribution with mean vector $\mathbf{Q}$ and covariance matrix Σ . Now if $T_1, \dots, T_k$ are independent $N(0, \frac{1}{2})$, then we can represent $Z_1, \dots, Z_{k-1}$ by $Z_i = T_k - T_i, i=1, \dots, k-1$. After conditioning on T_k , we see that the probability on the right of (5.2) is exactly that in (5.1).

6. Asymptotic Efficiencies

Consider two sequential selection procedures for the problem as stated in Section 1. Denote by N_T and N_S the stopping variables of the two procedures T and S respectively and by P_T and P_S the probabilities of a correct selection.

We define the asymptotic efficiency as $\Delta^* \rightarrow 0$ of procedure T relative to procedure S by

$$e(T, S) = \lim_{\Delta^* \rightarrow 0} (EN_S)/(EN_T)$$

provided the limit exists and $\liminf_{\Delta^* \rightarrow 0} P_T \geq P^*, \liminf_{\Delta^* \rightarrow 0} P_S \geq P^*$ for all $\omega \in \Omega$ and all $F \in H_0(J)$.

Denote by R the procedure of Robbins, Sobel and Starr (1968) and by G the procedure of section 5. Then it follows from (1.1) and theorem 5.1 that

$$e(R, G) = [A/2\sigma B(F)]^2$$

provided σ^2 (the variance of F) is finite. This efficiency is the same as the Pitman efficiency of the t-test with respect to the general one-sample rank order test. If $J(u) = \Phi^{-1}((1+u)/2)$, then it follows from Chernoff and Savage (1958) that $e(R, G) \leq 1$ for all

distribution functions F , with equality only when F is $N(0, \sigma^2)$.

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